

CURRICULUM VITAE

PERSONAL DETAILS

Date of Birth / Nationality 08.04.2000 / Indian
Marital Status / Gender Single / Male
Professional Profiles [LinkedIn](#) / [XING](#) / [Portfolio](#)



WORK EXPERIENCE

- 08.2025 – 01.2026

Mercedes-Benz AG, Stuttgart, Germany
Working Student – Battery Development, Algorithms & Applications.
 - Collaborated in fundamental research to expand and further develop the voltage-relaxation model for lithium-ion batteries with the goal of deepening understanding of battery behaviour.
 - Explored machine learning approaches for predictive analysis of battery performance, including practical experience in PyTorch and TensorFlow, as well as exploration of AI/ML models and hybrid (e.g., grey-box) modelling approaches for battery performance prediction.
 - Developed an anomaly dataset / anomaly database based on experimental battery measurement data to support future model training and validation processes.
- 12.2024 – 07.2025

Mercedes-Benz AG, Stuttgart, Germany
Master’s Thesis: Modelling and Prediction of Voltage Relaxation Behaviour in Lithium-Ion Batteries. (Grade: 1.0)
 - Conducted in-depth research on various relaxation models for electrochemical systems.
 - Developed an advanced relaxation model for lithium-ion batteries (LIBs) using modern Python frameworks.
 - Implemented an adaptive optimization algorithm based on experimental data, improving model accuracy by approximately 80–85%.
 - Applied and evaluated the gradient-based CylPOPT method to enhance model validation and overall performance.
 - Significantly improved state-of-charge (SOC) estimation accuracy, enabling more reliable and predictive battery performance assessment.
- 02.2024 – 11.2024

Mercedes-Benz AG, Stuttgart, Germany
Working Student in High-Voltage Battery Development – Systems Pre-Development.
 - Developed and implemented MATLAB scripts for automated analysis and visualization of measurement data, reducing manual workload by approximately 95–98%.
 - Assisted in the setup of 3D simulation models using CST Studio Suite.
 - Performed impedance measurements and created operating manuals for AC/DC magnetometers.
 - Analysed measurement data from relays and contactors to derive optimization measures.
 - Researched and compared methods for determining components magnetic permeability.
- 11.2023 – 01.2024

ElringKlinger AG, Neuffen, Stuttgart, Germany
Working Student in Battery Development – Battery System Testing & Validation.
 - Tested battery components and cells, including the creation and execution of test cases.
 - Evaluated measurement results using X-Frame and DIAdem.
 - Contributed to the hardware design of ECUs and microcontrollers.
- 05.2023 – 11.2023

ElringKlinger AG, Neuffen, Stuttgart, Deutschland
Intern: Battery System Testing & Validation.
 - Tested and validated battery systems, ECUs and drives to ensure performance and safety standards.
 - Performed impedance measurements on aluminium foils around the battery.
 - Created test protocols and documented test results.

EDUCATION

03.2022 – 08.2025	Hochschule Stralsund, Germany Master of Engineering – Renewable Energies & E-Mobility (Final Grade: 2.3) Key modules: Vehicle Management Systems, Vehicle Simulation & Test Driving, Systems Theory, Hydrogen Technology, Control of Electrical Drives.
08.2018 - 06.2021	University of Mumbai, India Bachelor of Engineering- Electrical Engineering (Final Grade: 1.3). Key modules: Power Systems, Electrical Networks, Electrical Machines, Power Electronics, Drives & Control, Environmental Management.
06.2015 - 05.2018	Maharashtra State Board of Technical Education (MSBTE), India Diploma in Electrical Engineering (Final Grade: 1.9). Key modules: Fundamentals of Electrical Engineering, Electrical Circuits & Networks, AC Machines, Power Electronics, Microcontrollers & Applications.

PROJECTS

12.2020 - 05.2021	Bachelor's Thesis Mumbai University, India Topic: Electromagnetic Space Shuttle Launch System (Research Based). Developed a linear electromagnetic accelerator to achieve high accelerations with minimal erosion wear in the rail/armature area. [IJERT]
08.2017 - 04.2018	Diploma Thesis MSBTE Mumbai, India Topic: Speed Control of a DC Motor using ARDUINO PWM Technique. - Designed a reliable system for constant speed control of a DC motor under variable load conditions. - Applied PWM technique with IGBT for power control and data logging. [GitHub]

SKILLS

Language Skills	German – B2.1 (actively improving) English – Fluent, written and spoken (C1) Marathi – Mother Tongue Hindi – Business proficient
Software Skills	Professional: MATLAB/Simulink, Python, VS Code, GitHub, Confluence, MS Office, GitHub Copilot Intermediate: Pandas, NumPy, PyTorch, TensorFlow, Git, Machine Learning (AI/ML), PREEvision, CAN, CANoe, CANalyzer, C, LT Spice, AUTOSAR, Impedance & Vector Analysis. Elementary: C++, Object-Oriented Programming.
Special Achievements	Second place in Bachelor of Engineering – Electrical Engineering, awarded a medal for outstanding academic performance.
Extra-Curricular Activities	01.2025 Systems Engineering Workshop – Mercedes-Benz AG, Stuttgart 1. Successfully completed an intensive four-day Systems Engineering course. 2. Acquired methods in requirements analysis, system architecture, integration and validation. 3. Developed a use-case analysis and system function specifications for HV components. 10.2019 Students Solar Ambassadors Workshop – IIT Mumbai and AIKTC, Mumbai 1. Student head in-charge for Solar Ambassadors Workshop conducted by AIKTC's Electrical Department organized globally by the Indian Institute of Technology, Bombay.
Interests	Chess Cooking Travelling.